Oxford Prospects and Global Development Institute

Course: STEM: Maths, Physics, Computer Science and Engineering

Session Title: Smart Cities after the Pandemic

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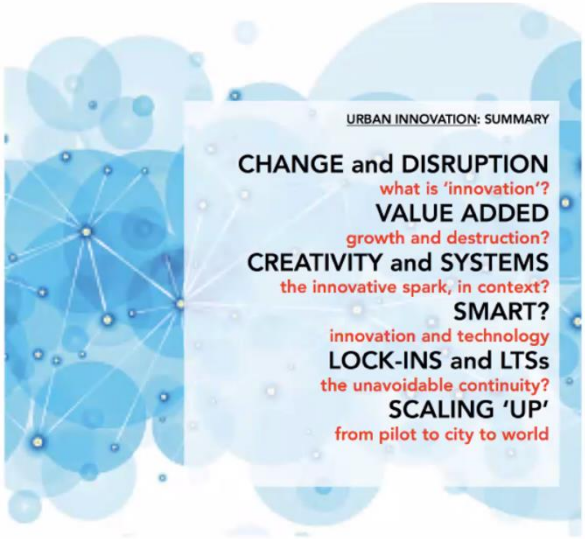
Date: 16/02/2022

Learning Objectives:

- Cities and smart cities
- The question of innovation and smart innovation in cities prior, and during, the Covid-19 Pandemic
- Feedback and questions from the group

Outline:

Urban innovation in the midst of a crisis



'Smart City' is a well-known phrase. Over the last two decades, the terminology about smart innovation has thrived. There are quite a few terminologies, but they sit in two main camps. When you study these things, you also come across questions of experimentation in cities, using the city as a laboratory. It's also a very political story.

Main Points:

Innovation has a very basic element about how cities have confronted crisis e.g., the hurricane in New York caused the centre to go completely dark and shut down. Some engineers provided a basic phone charging service. Why that example? Lots of urban innovation comes prompted by the need of crisis, they address basic services.

Innovation takes two phases, there is a brighter and darker side to it.

Change and disruption

- There has been an explosion of digitally enhanced forms of urban services. E.g., Deliveroo/Uber – a crowd-based platform
- These are sometimes not regulated and are handled in the same way as different services such as the treatment of Uber like a taxi service
- These innovations come and disrupt things and experiment with the cities

They're smart for whom?

- Many are not workable for generations who are not comfortable with them e.g., older generations and the concept of a touch screen or the idea of voice activated devices
- Smart cities have to go hand in hand with smart users

Problems and challenges

- Often, we learn by doing
- The question is – as you work on these technologies, if we do not provide a manual or instructions, how do users get to use and benefit from these innovations?
- Problems have been a basis of urban planning (Rittel and Webber)
- A key thing in the 70s was 'wicked problems' – our presence in the problem makes it different in itself e.g., a traffic control system may bring about more pedestrians – you can never really solve the problem, but you adapt and shift the problem
- 'Super wicked problems' (Levin, Lazarus, Bernstein)

Lock-ins

- Often when technology is built, we 'lock into' them a particular way of doing things
- E.g., your laptop has the "QWERTY" keyboard – why? Letters were scattered like that on typewriters, so people typed slower. The reason was a simple technological problem: faster typing would mean too much ink on the page.
- Innovations have built-in lock-ins in societies

Materiality and spaces

- Don't forget that large, global technical systems underpin specific innovations

Everyday technologies

- Many of these technologies emerge from everyday innovators

Cities as innovation ecosystems

- In themselves cities are very complex systems
- For example, we have institutions for knowledge within cities

Innovation, information and the 'infodemic'

- April 2020, WHO said "we're living in an infodemic" – a crisis of information – the power of information is a major challenge to the way people have addressed the pandemic
- The power of misinformation and rumours. E.g., pandemic side effects – people tweeted that the canals of Venice had gone clear, and swans had returned to canals. However, some of the pictures shared weren't located in Venice. The power of that information created certain imaginations and it affects the ways people act.
- It immediately created a system of information against information
- But this is not novelty. In the 1900s, London was in the midst of a cholera outbreak. In the context of lockdown, one piece of investigative journalism was looking at data and plotting it on a map. This was John Snow – correlations with cholera. Snow realised the water pump on Broad Street had more cases of cholera. This led to the spur which is

now the importance of using information to drive change in cities – same thing we're doing now in contact tracing and proximity alerts for Covid-19

- There has been a return of old practices

Is digital connectivity as effective in changing the way we imagine cities?

Benefits of forced experimentation (through crisis)

- During the two-day strike where the London tube stopped, people just changed their habits. People thought about their habit in a different way, and they learned in crisis which is also something that happened in the pandemic
- Learning and not having an option simply makes you learn new efficient ways to do things

Designing choice

- Don't underestimate the power of forced experimentation
- Remember that it is crucial to design choice – e.g., a major problem in the pandemic was having access to green space

Remembering the back of the shop

- How do our innovations avoid lock-ins and provide choice to people?
- What gets forgotten is the producers. There is a large world of production behind the consumption, and producers also need to be remembered
- E.g., cleaners, bus drivers… those that maintain the systems

Conclusions:

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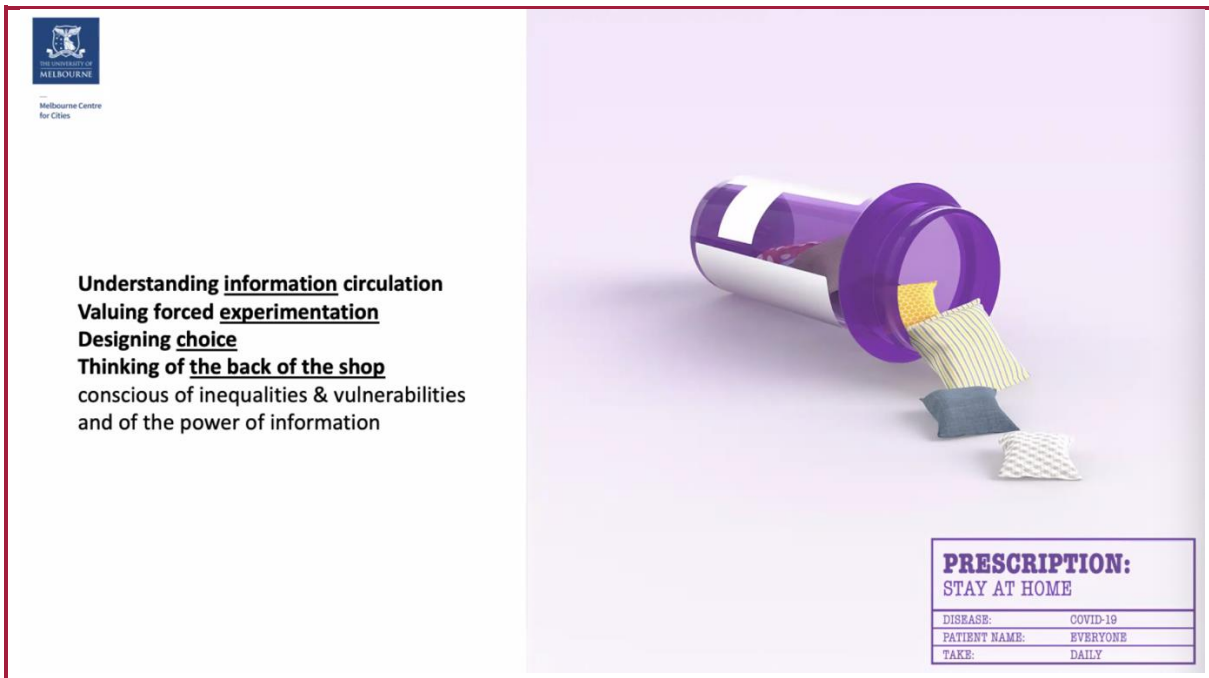

- Information circulation is as critical as innovation
- Recognise the value of forced experimentation, how forced changes can be positively leveraged
- Be aware of lock-ins
- We need to remember to design choice and be conscious of inequalities and vulnerabilities – think about the users and how it may be disruptive to certain generations

Q&A

Question:
I find our course uses many conventional ways to analyse cities. Are smart cities related more to computer sciences? What kind of skills do I need?

Answer:
Don't feel like you're by yourself, this is a common problem. It's reductive to teach smart cities as just computer sciences. Some basic coding and GIS really help, but some of the most useful elements in critical design thinking is the same for people who design furniture and phones. They teach a lot about materials and fabrication. Learning fashion and marketing. Really good training in urban innovation needs to go hand in hand with design thinking, creating, creating a cosmopolitan perception. Think about copying models… what's happening in other cities, would that work in my city?

Question:
What new technology do we need to learn about smart cities?

Answer:

Civil engineering is also a lot about paying attention to social and digital design. E.g., the PEARL lab – mobility in London. Civil engineering isn't just about building, it's also about designing.

Question:
What influence will smart cities take, when we design many choices, to interpersonal relationships? Will it be too convenient for individuals, will there be lack of conversations?

Answer:
It's a great question, and it's not easy to solve. There are two sides of the coin. The positive side is that the connectivity has allowed people to rely on their community during crisis. E.g., Discovery of neighbourhood friends during lockdowns and the thriving of localised groups. The flip side, the technology doesn't eclipse social relations. There are massive generational gaps of how people use the technology. The key problem is the use of what we make and how the technology is marketed. E.g., Dubai metro was marketed for tourists and visitors, but the major use was for medium to low-income migrants.

Question:
Is innovation easier to be copied, as its easier to be known?

Answer:
Don't be too precious about the process of innovation. A tangible example is Apple. They went out to make phones, it ended up being a computer. Everyone copied it. The purpose of design isn't to design solutions, it's to change problems and people will come and copy. This is exactly how innovation works. The problem is when we copy without paying attention to the original conditions that that innovation was put into in the first place. Critical copying is common. Often it doesn't work because they don't understand the context.

Question:
I have a concern about the older people, how can they get adapted to the smart cities?

Answer:
It's a concern that very few smart innovators have unfortunately. It's a question of thinking about what your own grandma might understand. With that perspective, innovation can have multiple users in the same device. E.g., assisted systems, tactile systems, or leapfrog innovations relying on audio rather than just tactile. Good design thinking teaches you this. One of the basic problems is money. People can't afford new innovations.

Question:
People's movement outdoors, how can we solve this problem?

Answer:
I don't think movement is the issue. It is the means by which those people move. We are often tackling the wrong problem. The key question is understanding the problem.

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Key concept: Planetary urbanisation. There is really no outside to the city. There are degrees of urbanisation. E.g., in the countryside. Some of the small towns are actually some of the most innovated places.
Further Reading:
- Problems as the basis of urban planning (Rittel and Webber) - 'Super Wicked Problems' (Levin, Lazarus, Bernstein)